Depositing More Should Not Earn Less

Supplier income monotonicity and rate curve design in pooled lending markets

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Abstract

In utilization dependent interest rate models, adding liquidity can reduce a supplier's total interest income even as the supplier's balance increases. A supplier holding a fraction α of final pool supply keeps income nondecreasing with a marginal deposit exactly when $\alpha(1+\eta) \leq 1$, where η is the elasticity of the borrow rate with respect to utilization. When the condition fails, the supplier can instead raise its income by withdrawing, which pushes utilization further up. A protocol turns this into a design constraint by choosing, at each utilization, the largest supplier share it intends to protect. That choice caps the elasticity, and integrating the cap describes every admissible curve. Protecting the same share at every utilization gives a power law boundary, and protecting every fresh depositor gives a hyperbolic one. On a multiple kink USDT market whose parameters are drawn from a deployed configuration, the constraint does not determine the level of interest income. A compliant floor can reduce earnings, while a repair that only raises rates preserves them at a cost to borrowers.

1 Introduction

Pooled lending protocols commonly determine interest rates as a function of utilization [1,7]. A standard design increases the borrow rate gradually below an optimal utilization level and much more aggressively above it. The steep branch improves incentives to restore liquidity under stress, but it also creates a supplier price impact problem: an additional deposit lowers utilization, which lowers the borrow rate, and the lower rate applies to the supplier's entire balance.

Consequently, a supplier may be better off withholding capital, or withdrawing it. On a lending market at high utilization, which is when the pool most needs supply, one of us found that depositing only part of the available wallet balance earned more than depositing all of it.

Existing work on these curves treats lenders as a group or the pool as a whole: it compares the rate rules of deployed protocols [6], models lending pools formally [3], or designs rates that maximize expected lender wealth [4] or keep utilization stable [2]. Closest to the question here, Baude et al. [5] account for the effect of a borrower's own position size on its borrowing cost. We study the corresponding question for a single supplier.

The role of this paper is to give an exact condition under which this cannot happen, to describe the rate curves that satisfy it, and to separate what such a guarantee promises from

what it costs. Section 2 works the failure through in numbers. Sections 3–5 derive the income condition and the admissible family of rate curves it induces. Sections 6 and 7 develop two protection guarantees and show that both are instances of the same construction, and Section 8 gives a power law repair. Sections 9 and 10 take a market with several kinks and separate protection from pricing, and Section 11 states the check a protocol would implement. Detailed constructions, sensitivity studies and the smooth extension are collected in the appendices.

Most of the results are formalized in Lean 4. The Lean library, the analysis code and its data are available at `https://github.com/0xahzam/supplier-monotonicity`, with a record of which statements are covered.

2 The Failure Mode

Consider a market with optimal utilization $O = 80\%$, a borrow rate of $R_o = 5\%$ at that point, and a maximum borrow rate of $R_{\max} = 100\%$ at full utilization, connected by straight lines. This is a standard kinked curve [1], and the change in slope at the kink is large but not unusual.

Let the market hold \$10 million of supply against \$8.3 million of borrowings, so utilization is 83%. The market is above its own optimal level, so the protocol wants more supply. A supplier that already holds \$1 million, 10% of the pool, adds \$375,000, exactly enough to return the market to optimal utilization. Table 1 shows the result. The balance grows by 37.5% and the income falls by 66%, because the lower rate now applies to the whole balance. Depositing past the kink recovers only part of the loss, and as Figure 1 shows, the income never returns to where it started. The same supplier does better by withdrawing about \$302,000, which moves utilization further from the level the protocol is trying to restore.

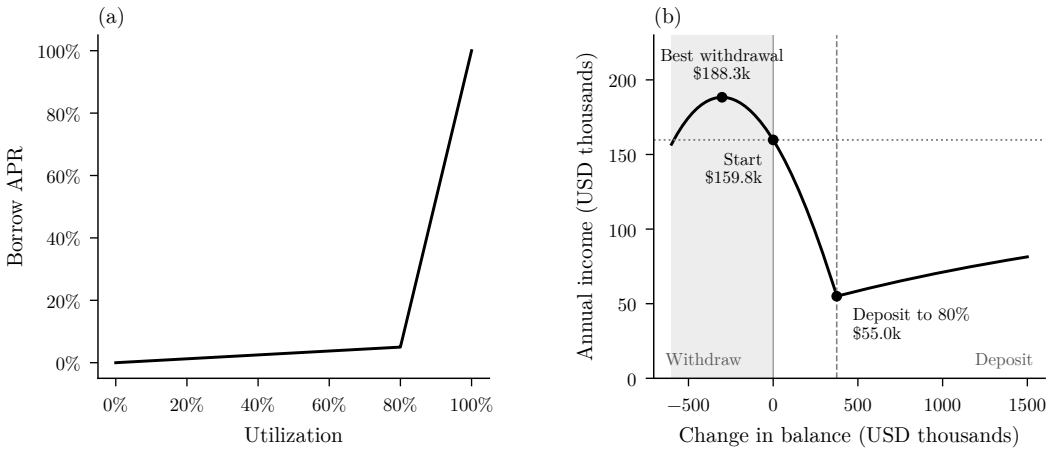


Figure 1: (a) The illustrative kinked borrow rate. (b) Annualized income of a supplier initially holding 10% of a \$10 million pool, with \$8.3 million of fixed borrowing, against the change in its balance. Withdrawals are shaded. The dotted line is the starting income, and the dashed line marks the deposit that returns utilization to 80%. The labels give income at the start, after that deposit, and at the income-maximizing withdrawal.

Table 1: The 10% supplier at three balances. Borrowings stay at $8.3 million. Income is annualized at the resulting state. Values in the last row are rounded.

	Balance	Total supply	U	Borrow rate	Supply rate	Income
Start	$1,000,000	$10,000,000	83%	19.25%	15.98%	$159,775
Deposit $375,000	$1,375,000	$10,375,000	80%	5%	4%	$55,000
Withdraw $302,000	$698,000	$9,698,000	85.6%	31.5%	27.0%	$188,300

Two features of this example matter later. The loss comes from the sensitivity of the rate, not from the size of the deposit. The worst point is the kink, not full utilization. Both follow from the condition derived below, and neither is visible from the slope of the curve alone.

3 Model

Let

- T denote total supplied tokens after the supplier's transaction,
- B denote total borrowings,
- b denote the supplier's resulting deposited balance,
- $U = B/T$ denote utilization,
- $R_b(U)$ denote the borrow rate at utilization U, and
- $R_s(U)$ denote the supply rate.

We assume the rate is nondecreasing, continuous and piecewise continuously differentiable, with $R_b(U) > 0$ for $U > 0$. The formulas below apply inside each smooth segment. At a kink, where the slope changes, a deposit lowers utilization, so the slope that matters is the one of the segment below. A zero base rate at $U = 0$ is permitted unless a later guarantee rules it out.

Ignoring fees, aggregate borrowers pay $BR_b(U)$ in annualized interest. Distributed across T supplied tokens, the supply rate is therefore

$$R_s(U) = \frac{BR_b(U)}{T} = UR_b(U) \tag{3.1}$$

and the supplier's annualized interest income is

$$I = bR_s(U) = bUR_b(U) \tag{3.2}$$

Income here is the annual rate at the current state, not interest already earned. Protocols usually keep a fixed share of borrower interest, often called the reserve factor. A constant share $f < 1$ scales supplier income and its derivative by $1 - f$ at every utilization, so it does not change the sign condition, and the analysis can ignore it. A fee that varies with utilization would have to be included in the derivative.

Write the supplier's share of final pool supply as

$$\alpha = \frac{b}{T} \tag{3.3}$$

The share is measured after the deposit, at the supplier's resulting state. Equivalently $I = \alpha B R_b(U)$, so the supplier receives the share α of all borrower interest.

Consider an infinitesimal increase $d\delta$ in the same supplier's deposit, holding B fixed during the transaction. Then

$$\frac{dU}{d\delta} = \frac{d}{d\delta}\left(\frac{B}{T}\right) = -\frac{B}{T^2} = -\frac{U}{T} \tag{3.4}$$

Borrowings respond to rates, but not within the transaction. A deposit settles against the current borrowings, and any borrowing that the lower rate attracts arrives later and offsets part of the loss. The condition describes that immediate effect, which is the one a supplier sees when deciding whether to deposit. How far borrowers respond afterwards is outside the claim, as in Section 10.

4 Supplier Income Monotonicity

Differentiating $I = bUR_b(U)$ with respect to the additional deposit gives

$$\begin{aligned}
\frac{dI}{d\delta} &= \frac{d}{d\delta}\left[bUR_b(U)\right] \\
&= UR_b(U) + b\left[R_b(U) + UR_b'(U)\right]\frac{dU}{d\delta} \\
&= UR_b(U) - \frac{bU}{T}\left[R_b(U) + UR_b'(U)\right] \\
&= UR_b(U)\left[1 - \frac{b}{T}\left(1 + \frac{UR_b'(U)}{R_b(U)}\right)\right]
\end{aligned} \tag{4.1}$$

Proposition 1 (Marginal supplier income)**.** *For a marginal deposit made while borrowings are fixed,*

$$\boxed{\frac{dI}{d\delta} = UR_b(U)\left[1 - \alpha\left(1 + \frac{UR_b'(U)}{R_b(U)}\right)\right]} \tag{4.2}$$

Two competing effects sit inside that bracket: the marginal dollar earns the current supply rate $UR_b(U)$, while the fall in utilization it causes lowers the rate applied to everything the supplier already holds. The size of the second effect is measured by $UR_b'(U)/R_b(U)$, the elasticity of the borrow rate with respect to utilization,

$$\boxed{\eta(U) = \frac{U}{R_b(U)}\frac{dR_b}{dU} = \frac{UR_b'(U)}{R_b(U)} = \frac{d\ln R_b}{d\ln U}} \tag{4.3}$$

The logarithmic form on the right is what Section 5 works with. Substituting it into the income derivative gives the identity this paper is built on,

$$\boxed{\frac{dI}{d\delta} = UR_b(U)\left[1 - \alpha(1 + \eta(U))\right]} \tag{4.4}$$

Since the prefactor is positive,

$$\frac{dI}{d\delta} \geq 0 \quad \Longleftrightarrow \quad \alpha(1+\eta) \leq 1 \tag{4.5}$$

and the largest supplier share whose marginal deposit remains income nondecreasing is

$$\boxed{\alpha_{\text{safe}}(U) = \frac{1}{1+\eta(U)}} \tag{4.6}$$

An elasticity of 9 protects a 10% supplier, 19 protects 5%, and 99 protects 1%. Whether a curve is too steep for a given supplier therefore depends on its elasticity, the percentage change in rate per percentage change in utilization, and not on its slope.

A finite deposit is a chain of marginal ones. If the condition holds at every point along the way, income at the end is at least income at the start. This also holds across kinks, because income is continuous there.

The condition also explains both features of Section 2. Above the kink the slope is $m = (R_{\max} - R_o)/(1 - O) = 4.75$, and on a straight segment $R_b' = m$. At the starting utilization of 83% the borrow rate is 19.25%, so $\eta = 0.83 \times 4.75/0.1925 \approx 20.5$ and $\alpha(1+\eta) \approx 0.1 \times 21.5 \approx 2.15$. That is more than twice the limit, so the deposit loses money, as Section 2 found. On the steep branch elasticity is $Um/(R_o + m(U - O))$. It is largest just above the kink, where the rate is lowest. There $\eta = Om/R_o = 76$, so only suppliers below $1/77 \approx 1.3\%$ of the pool are protected. At full utilization it falls to $m/R_{\max} = 4.75$, because the same slope moves a high rate by a smaller percentage. The steep branch is therefore most dangerous at its foot rather than its top. Below the kink this curve has $\eta = 1$, so the condition holds again for any share up to one half and income starts to rise. That is why the income trough in Figure 1 sits exactly at the kink.

Read in the other direction, the same condition prices a withdrawal. Wherever $\alpha(1+\eta) > 1$ the derivative in Equation 4.4 is negative, so removing liquidity raises the supplier's income. Withdrawing lowers the supplier's share and raises utilization, which on the steep branch lowers elasticity, so $\alpha(1+\eta)$ falls until income peaks where it reaches 1. In Section 2 this is the withdrawal of about \$302,000, at which $\alpha(1+\eta) = 1$ exactly. So the supplier the pool most needs earns more by withdrawing, and withdrawing pushes utilization further above O.

5 The Admissible Family

The condition $\alpha(1+\eta) \leq 1$ describes a supplier rather than a curve. A protocol cannot control α, but it can decide whose deposits to keep profitable. Write $\bar{\alpha}(U)$ for the largest supplier share the protocol intends to protect when the resulting utilization is U, and call this a *protection profile*. Imposing the condition at $\alpha = \bar{\alpha}(U)$ gives $1 + \eta \leq 1/\bar{\alpha}$, a ceiling on elasticity that we call the *elasticity envelope* E induced by the profile. The ceiling also covers every smaller supplier, because $\alpha(1+\eta)$ grows with α,

$$\boxed{\eta(U) \leq E(U) \quad \text{where} \quad E(U) = \frac{1 - \bar{\alpha}(U)}{\bar{\alpha}(U)} \qquad \forall U} \tag{5.1}$$

Protecting more of the pool forces a lower envelope: a target of $\bar{\alpha} = \frac{1}{2}$ allows nothing beyond unit elasticity, $\bar{\alpha} = 5\%$ allows 19, and as $\bar{\alpha} \to 0$ the envelope grows without bound and the constraint disappears.

During a deposit the supplier's share only grows. Under a constant $\bar{\alpha}$ it is therefore enough to check the final share. A profile that varies with U must be checked along the path.

Since elasticity is a logarithmic derivative (Equation 4.3), writing the constraint as

$$\frac{d \ln R_b}{d \ln U} \leq E(U) \tag{5.2}$$

and integrating it bounds the curve itself. We say that R_b *honors* the profile $\bar{\alpha}$ at U when $\eta(U) \leq E(U)$.

Theorem 2 (The admissible family)**.** *Let $0 < U_a < U_b < 1$, let R_b be positive and locally absolutely continuous, and let $E(U)/U$ be integrable on the interval. Then R_b honors the profile $\bar{\alpha}$ almost everywhere on $[U_a, U_b]$ if and only if*

$$\boxed{R_b(U) \leq R_b(U_a) \exp\left(\int_{U_a}^{U} \frac{E(v)}{v}\, dv \right)} \tag{5.3}$$

holds for every $U \in [U_a, U_b]$ and on every subinterval. Equality throughout characterizes the boundary curve *of the profile, the solution of $U R_b'(U) = E(U) R_b(U)$ anchored at $(U_a, R_b(U_a))$. Endpoint statements follow by limits where those limits exist.*

Proof. If $\eta \leq E$, then $R_b'/R_b \leq E/U$. The left side is the derivative of $\ln R_b$, so integrating from U_a to U and exponentiating gives the bound. Conversely, apply the bound to a short interval $[U, U + h]$, take logarithms and divide by h. The left side becomes the average slope of $\ln R_b$ over the interval, and the right side the average of $E(v)/v$. As $h \to 0$ they tend to $R_b'(U)/R_b(U)$ and $E(U)/U$, which gives $\eta(U) \leq E(U)$. These limits exist at every point outside a set of zero length, which is why the result holds almost everywhere. For the piecewise smooth curves used here they exist inside every segment. □

The next sections develop a constant share target and protection for every fresh depositor, and Appendix A illustrates their boundary curves.

6 Fixed Supplier Share Protection

The simplest policy is to protect the same maximum supplier share at every utilization. The profile is then a constant $\bar{\alpha}$, so the envelope in Equation 5.1 is a constant too,

$$\boxed{K = \frac{1 - \bar{\alpha}}{\bar{\alpha}}} \tag{6.1}$$

and the design rule is $\eta(U) \leq K$ everywhere.

With $E \equiv K$ in Theorem 2, the boundary anchored at the kink (O, R_o) is

$$\boxed{R_K(U) = R_o \left(\frac{U}{O} \right)^K} \tag{6.2}$$

This power law is the equality case of Equation 5.3, so its elasticity is identically K. The ceiling K comes from choosing a flat target and is not a general restriction on interest rate models.

6.1 The Endpoint Impossibility Bound

A rate curve must still reach $R_{\max}$ at full utilization, and fixed endpoints therefore limit how small the worst case elasticity can be. Integrating Equation 4.3 from O to 1 shows that any positive differentiable curve connecting (O, R_o) to $(1, R_{\max})$ satisfies

$$\ln \frac{R_{\max}}{R_o} = \int_O^1 \eta(U)\, d\ln U \tag{6.3}$$

so if $\eta \le K$ everywhere then $\ln(R_{\max}/R_o) \le K \ln(1/O)$.

Theorem 3 (Endpoint impossibility bound)**.** *Any positive, absolutely continuous borrow rate curve with endpoints (O, R_o) and $(1, R_{\max})$ has essential supremum of elasticity at least*

$$\boxed{K^* = \frac{\ln(R_{\max}/R_o)}{\ln(1/O)}} \tag{6.4}$$

so an elasticity ceiling $K < K^*$ is infeasible whatever the shape of the curve.

The bound is tight, since the power law with $K = K^*$ reaches $R_{\max}$ exactly at $U = 1$. Appendix E shows that if the curve must also have a continuous derivative, no curve attains K^*.

Since protected share is $1/(1+K)$, the endpoints also cap the guarantee a protocol can offer without moving them,

$$\boxed{\bar{\alpha}_{\max} = \frac{1}{1+K^*}} \tag{6.5}$$

Optimal utilization, the rate at optimal utilization, the maximum rate, and the supplier protection target therefore cannot be chosen independently of one another. For the illustrative inputs $O = 80\%$, $R_o = 5\%$ and $R_{\max} = 100\%$ this gives $K^* \approx 13.43$ and $\bar{\alpha}_{\max} \approx 6.93\%$, so no curve through those endpoints guarantees protection to a supplier holding more than about 7% of pool supply.

7 The Fresh Depositor Guarantee

A different policy is to protect anyone arriving with no existing position, whatever the state of the market when they arrive. Suppose such a supplier deposits x into a pool with initial supply S_0 and fixed borrowings B, and write

$$u_0 = \frac{B}{S_0} \qquad \text{and} \qquad u = \frac{B}{S_0 + x} \tag{7.1}$$

for utilization before and after. Their final pool share depends only on those two utilizations,

$$\alpha = \frac{x}{S_0 + x} = 1 - \frac{u}{u_0} \tag{7.2}$$

and substituting into $\alpha(1+\eta) \le 1$ gives the exact condition along a deposit path beginning at u_0,

$$\boxed{(u_0 - u)R_b'(u) \le R_b(u)} \tag{7.3}$$

For a nondecreasing rate the left side grows with u_0, so the most demanding starting point is $u_0 = 1$, where $\alpha = 1 - u$. Protecting every fresh depositor from every feasible start is therefore just another protection profile, and everything in Section 5 applies unchanged.

Theorem 4 (Global fresh depositor condition)**.** *Protecting a fresh depositor entering from any initial utilization is the protection profile*

$$\boxed{\bar{\alpha}(U) = 1 - U} \tag{7.4}$$

whose envelope is

$$\boxed{E(U) = \frac{U}{1-U}} \tag{7.5}$$

so the design rule is $\eta(u) \le u/(1-u)$, equivalently $(1-u)R_b'(u) \le R_b(u)$, and the boundary curve is

$$\boxed{R_b(U) = R_b(U_a)\,\frac{1-U_a}{1-U}} \tag{7.6}$$

Proof. Substituting $\bar{\alpha} = 1-U$ into Equation 5.1 gives $E = U/(1-U)$, and $\eta \le E$ rearranges to $(1-u)R_b' \le R_b$. Since $E(v)/v = 1/(1-v)$, the integral in Equation 5.3 evaluates to $\ln\left((1-U_a)/(1-U)\right)$, which gives the stated boundary. □

The profile protects half the pool at 50% utilization, a tenth of it at 90%, and only a hundredth at 99%. Promising little near full utilization is what lets the rule tolerate a steep emergency branch, and it is also the rule's main limitation, since it promises nothing to the larger suppliers a pool short of liquidity most needs. Figure 2 compares the two profiles and the envelopes they induce.

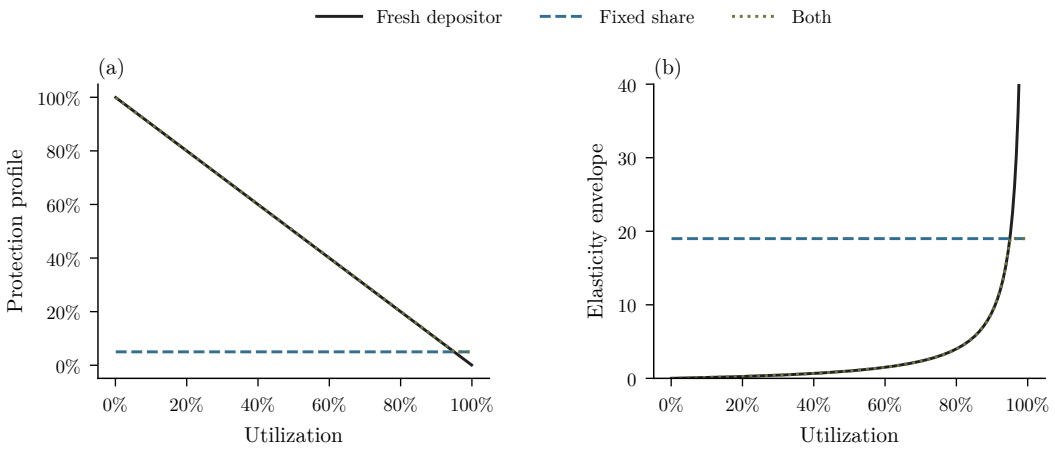


Figure 2: (a) Protection profiles and (b) their elasticity envelopes. The flat target is 5%. Imposing both guarantees takes the larger protection profile and smaller envelope. The envelopes cross at $U = K/(1+K) = 95\%$.

The two envelopes cross at $U = K/(1+K)$, which is 95% for $K = 19$. Below that point the fresh depositor rule is the *stricter* of the two and above it the flat target is, so neither guarantee implies the other.

Two further consequences separate this profile from a flat one. The first is that a fresh depositor guarantee cannot be offered by a curve through the origin. Applying Theorem 4 on $[U_a, U]$ and letting $U_a \to 0$ gives $R_b(U) \le R_b(0)/(1-U)$, that is

$$R_b(0) \ge R_b(U)\,(1-U) \qquad \forall U < 1 \tag{7.7}$$

so the base rate is bounded below by the whole curve and in particular is strictly positive. The standard below kink ramp fails for exactly this reason: being proportional to U it has $\eta \equiv 1$, while the envelope $U/(1-U)$ is below 1 for every $U < \frac{1}{2}$. The example curve in Appendix A, Table 2, starts at a positive rate.

The second is that a positive anchor below full utilization places no upper bound on the rate attainable at $U = 1$ when the shape between them is unrestricted. For a flat target the available rate budget $\int_O^1 K\, d\ln U$ is finite. Here the corresponding integral is $\ln((1-O)/(1-U))$, which diverges as $U \to 1$. A finite maximum above the anchor can therefore be reached by following the boundary and then holding the rate flat. Fixed kinks add restrictions, discussed in Section 9.

7.1 Piecewise Linear Form

Onchain curves are piecewise linear, and the guarantee takes an exact form on them. Take a nondecreasing segment from (u_L, R_L) to (u_H, R_H) with slope

$$m = \frac{R_H - R_L}{u_H - u_L} \tag{7.8}$$

so that $R_b(u) = R_L + m(u - u_L)$ inside it. The condition $\eta \le u/(1-u)$ is equivalent to $m(1-u) \le R_b(u)$, and since $R_b(u)/(1-u)$ increases along a nondecreasing segment, the binding point is the left endpoint. The segment is admissible exactly when $0 \le m \le R_L/(1-u_L)$, and substituting the slope turns this into a rule on adjacent kinks alone,

$$\boxed{\frac{R_H}{R_L} \le 1 + \frac{u_H - u_L}{1-u_L}} \tag{7.9}$$

If every segment of a continuous piecewise linear curve satisfies Equation 7.9, income stays nondecreasing even for a deposit that crosses several segments, and no condition linking one segment to the next is required. A flat clamp after the final kink preserves the condition as well.

Appendix A gives an admissible curve and follows a fresh deposit across its kinks, separating the sign of marginal income from the level of income.

8 A Minimal Change Construction

Return to a flat target and suppose a protocol already runs a single kink model

$$R_{\text{old}}(U) = \begin{cases} R_o \dfrac{U}{O}, & U \le O \\ R_o + m(U - O), & U > O \end{cases} \qquad m = \frac{R_{\max} - R_o}{1 - O} \tag{8.1}$$

Assume $K \ge 1$ and the upper branch has negative intercept, $R_o - mO < 0$, as in the illustrative curve. Below the optimum $\eta = 1$ and nothing needs to change. Above it $\eta_{\text{old}}(U) = Um/(R_o + m(U - O))$ is largest at the kink. For $K \ge K^*$, let U_c be the first utilization above O at which the power law boundary catches the original branch,

$$R_o \left(\frac{U_c}{O}\right)^K = R_o + m(U_c - O) \tag{8.2}$$

For a feasible K the natural construction follows the boundary only over that interval,

$$\boxed{R_{\text{new}}(U) = \begin{cases} R_o \dfrac{U}{O}, & U \le O \\ R_o \left(\dfrac{U}{O}\right)^K, & O < U < U_c \\ R_o + m(U - O), & U_c \le U \le 1 \end{cases}} \tag{8.3}$$

preserving the curve below the kink and after the rejoin. The replacement can extend into a region where the original elasticity is already below K, because the power law branch runs below the original and must still catch up with it. If the original curve is admissible everywhere, no replacement interval is needed. As $\bar{\alpha} \to 0$ the ceiling $K \to \infty$, and once K exceeds the maximum original elasticity,

$$\boxed{\bar{\alpha} \to 0 \quad \Longrightarrow \quad R_{\text{new}}(U) = R_{\text{old}}(U)} \tag{8.4}$$

so the construction is better read as a constraint imposed on an existing interest rate model than as a proposal to replace it.

Figure 3 compares borrow and supply rates for $O = 80\%$, $R_o = 5\%$, $R_{\max} = 100\%$ and $K = 19$. The branch changes the path between the fixed endpoints while keeping both the kink rate and the maximum rate.

Over its replacement interval this curve pays less than the baseline at the same utilization. The benefit is that adding capital cannot reduce whole position income while the supplier remains within the protected share. Appendix A gives a finite deposit example. Preserving existing income levels requires a different pricing choice, considered below.

9 Several Kinks and the Choice of Rates

Consider a USDT market, with parameters drawn from a deployed configuration, having a 0.5% minimum borrow APR, an 80% main kink at 5.25% APR, and a maximum borrow APR of 90%. Further kinks lie at 85%, 90%, 95%, 99% and 99.5% utilization, with rates specified in Appendix B. The example lets us distinguish the protection a curve provides from the price levels chosen to provide it.

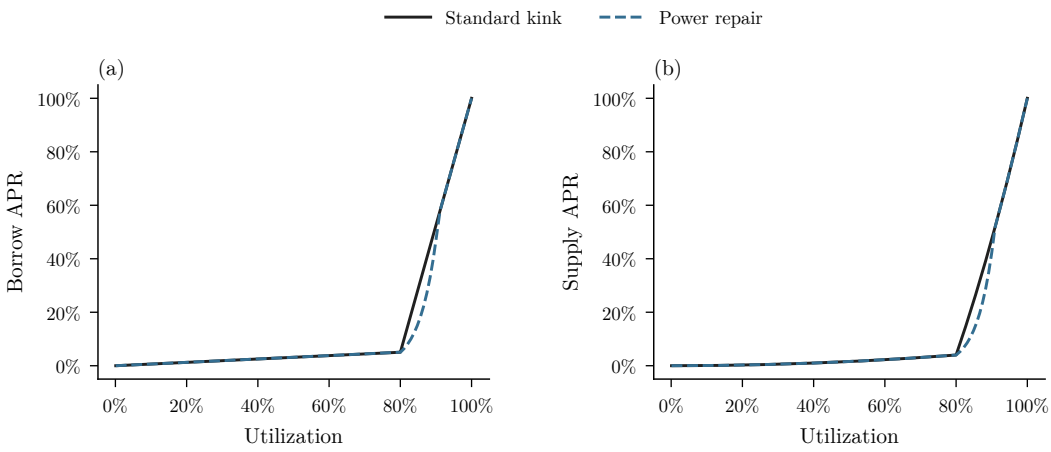


Figure 3: (a) Borrow and (b) supply APR before and after the power law repair with $K = 19$. The repaired branch rejoins at approximately 90.92% utilization. Supply APR is $UR_b(U)$, without fees.

To protect both fresh depositors and existing suppliers up to a fixed share, impose the stronger promise at each utilization:

$$\bar{\alpha}_{\text{both}}(U) = \max\{\bar{\alpha}, 1 - U\} \qquad E_{\text{both}}(U) = \min\left\{K, \frac{U}{1-U}\right\} \tag{9.1}$$

At a 5% fixed share, the fresh depositor rule binds below 95% utilization and the fixed share rule above it. A power law region alone does not establish both guarantees, because every retained segment must also respect the tighter envelope. Figure 4 shows the protected share and the marginal income of a fresh depositor under both guarantees.

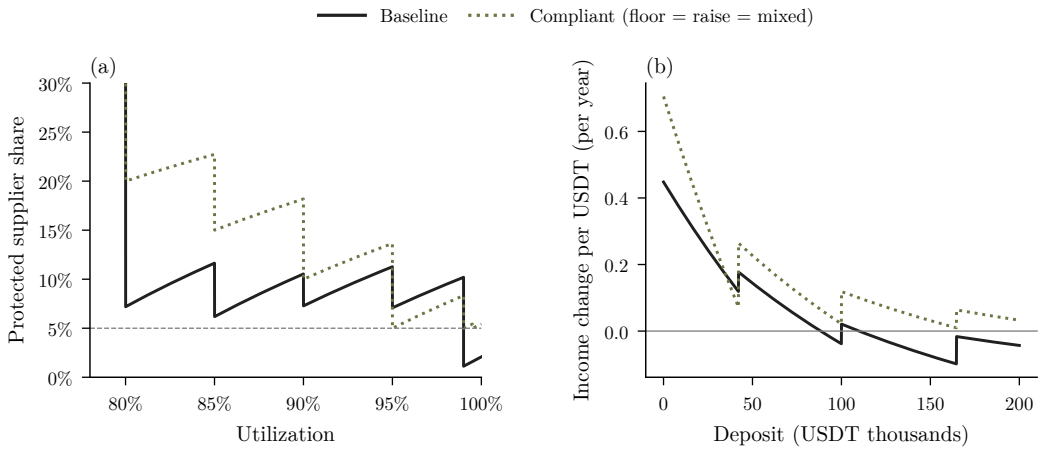


Figure 4: Both guarantees on the fixed kinks. (a) Actual protected share, with the 5% target marked. (b) Marginal whole position income for a fresh depositor entering a pool with 1 million USDT supply and 990,000 USDT of fixed borrowing. The floor, raise only and mixed curves coincide here. Their marginal income stays nonnegative across the crossed kinks, but the baseline's does not.

With kinks and maximum APR fixed, the segment constraints place a floor under preceding rates. Each guarantee caps the ratio between adjacent kink rates, $r_{i+1} \leq c_i r_i$. For the fresh depositor guarantee c_i is the right side of Equation 7.9, and Appendix B gives it for each guarantee. The floor is obtained by propagating the final rate backwards as $\ell_i = \ell_{i+1}/c_i$. Moving every rate to the floor can lower rates that were already acceptable. A repair that preserves them instead sets

$$r_n^+ = r_n^0 \qquad r_i^+ = \max\left\{r_i^0, \frac{r_{i+1}^+}{c_i}\right\} \tag{9.2}$$

where r_i^0 denotes the baseline. This is the smallest admissible repair that never lowers a rate, so it preserves supplier income under the same principal and utilization history at a cost to borrowers. Figure 5 compares these choices with a mixed repair that permits increases and decreases while minimizing weighted changes to kink rates. The recursions, objective and feasibility conditions are derived in Appendix B.

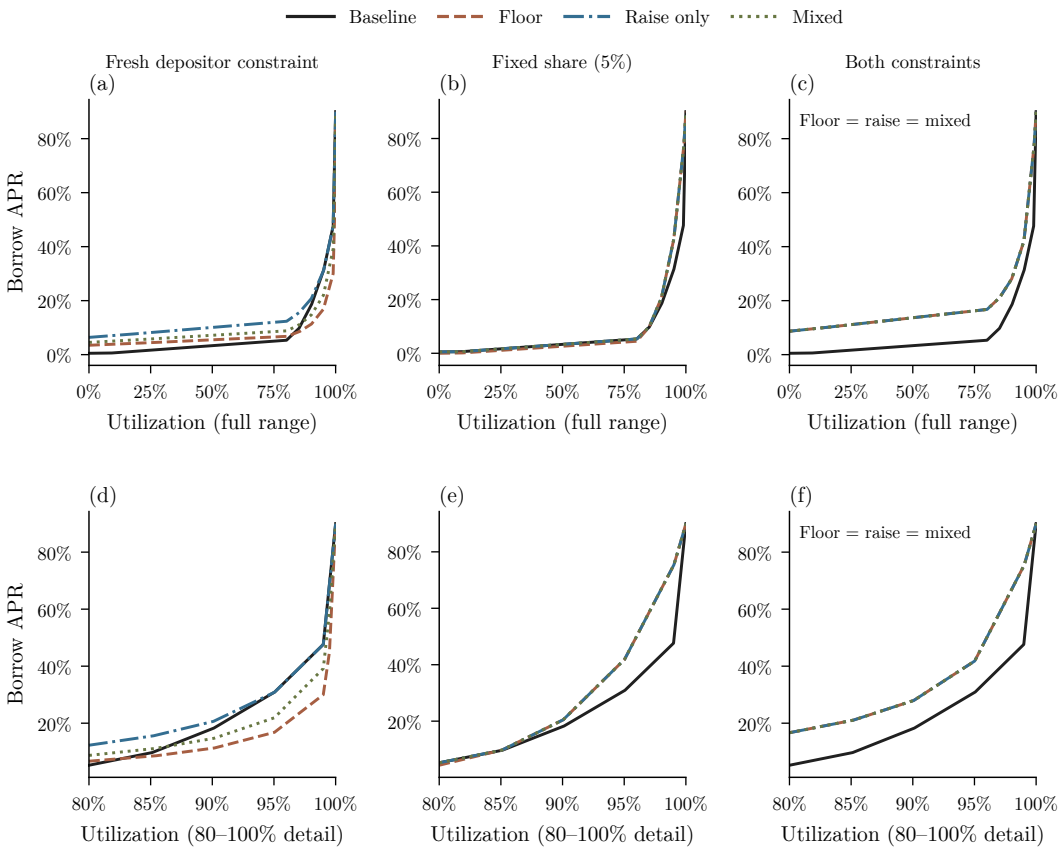


Figure 5: Baseline, compliant floor, raise only repair and mixed repair, at fixed kinks and 90% top APR. Columns impose the fresh depositor guarantee, the 5% fixed share guarantee, and both. Panels (a–c) show 0–100% utilization, and panels (d–f) show the same curves enlarged over 80–100%, with no change in parameters or scenario. Under both constraints the floor is already above the baseline, so the three repairs coincide.

The bound also depends on how the curve is built. On the original kinks, both guarantees with 5% fixed share protection and a 90% maximum require a base APR of at least 8.68%.

Allowing unrestricted curved regions reduces the combined bound to 1.70%. Requiring the analytic curve also to remain above the baseline gives approximately 1.85%. These are different constraints, not conflicting estimates. Appendix B derives the bounds and shows their dependence on kink spacing.

A similar restriction appears when inserting power laws around several kinks. A branch that only lowers rates after the late 99% kink reaches 57.65% at full utilization and cannot retain the original 90% maximum. Preserving that maximum requires changing rates earlier. The analytic constructions in Appendix B.3 locate these replacement regions directly and distinguish repairs that never raise rates from those that never lower them.

10 Income Levels and Accumulated Interest

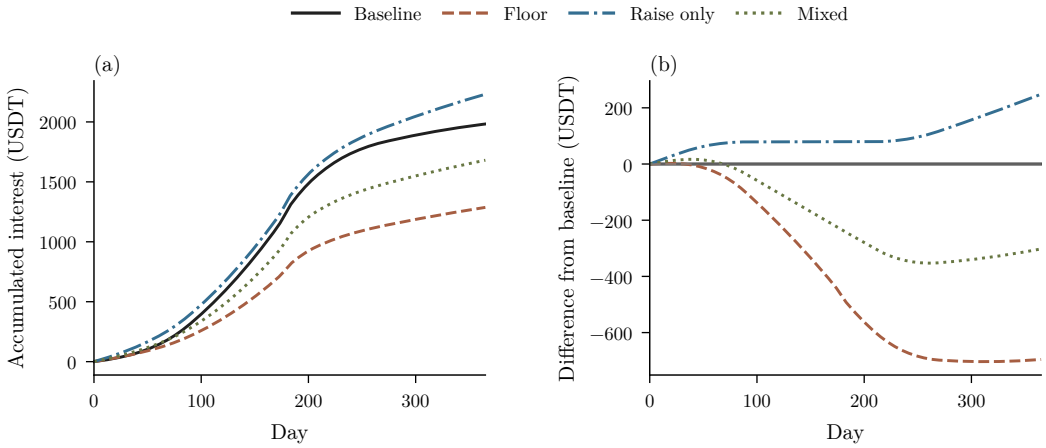


Figure 6: Fresh depositor guarantee only, on the stated single deposit history. (a) Accumulated interest and (b) difference from baseline. A compliant floor is not the same choice as a repair that preserves existing rates. Principal is excluded from interest.

Whether a depositor earns more over a holding period depends on the rates they encounter and the capital they hold at each point, rather than on the sign of the deposit derivative alone. For prescribed principal $b(t)$ and utilization $U(t)$, with a constant reserve factor f, accumulated interest is

$$A(H) = \int_0^H (1 - f) b(t) U(t) R_b(U(t))\, dt \tag{10.1}$$

where time is measured in years. Comparing two curves on identical histories gives

$$\Delta A(H) = \int_0^H (1 - f) b(t) U(t) \left[R_1(U(t)) - R_0(U(t)) \right] dt \tag{10.2}$$

A raise only repair therefore cannot reduce accumulated supplier interest under the same history, and a lower only repair cannot increase it. A mixed change depends on where the depositor spends time and how much capital is present there.

Each comparison below holds borrowings fixed and replays an identical utilization path under every curve, which isolates the effect of pricing. A curve charging borrowers more

would in practice face different borrowing and a different path, so the figures show what each curve would have paid on a given history, not how the market would respond, which needs a demand model such as those of [2, 4].

The examples use a constant 5% reserve factor on borrower interest, with interest paid out rather than compounded into the position. They evaluate the continuous curves and omit additional fees, incentives, losses and token rounding, so differences come entirely from the prescribed rate functions and market histories.

First take a 10,000 USDT depositor in a pool with 1 million USDT supply. They hold for one year while utilization moves linearly through 80%, 95%, 99.5%, 85% and 80% on days 0, 90, 180, 270 and 365. Under the fresh depositor guarantee alone, the floor pays about 1,287 USDT, the raise only repair 2,231, and the mixed repair 1,680, against 1,983 under the baseline. Figure 6 shows the accumulated interest over the year. All three repairs obey the same guarantee and keep the same maximum. Their different earnings come from the choice of rate levels.

Higher supplier earnings on an unchanged history are funded by borrower payments. Since this depositor owns 1% of the pool, each additional unit of aggregate borrower interest contributes 0.0095 units to the depositor after the reserve factor. The guarantee controls the response to adding capital, not whether the price level suits either side of the market.

Appendix C considers repeated deposits, stronger protection targets and different utilization exposures.

11 Checking the Constraint Onchain

Onchain implementations represent rate curves with linear segments rather than evaluating fractional powers or exponentials, and the flat share constraint can be checked directly in that form. For a segment from (u_i, r_i) to (u_{i+1}, r_{i+1}) with slope m_i,

$$R_b(U) = r_i + m_i(U - u_i) \qquad \eta(U) = \frac{U m_i}{R_b(U)} \tag{11.1}$$

and elasticity is monotone across a positive linear segment, so checking both endpoints bounds it over the whole segment. Writing $\Delta r = r_{i+1} - r_i$ and $\Delta U = u_{i+1} - u_i$, the condition at each endpoint (U, r) of the segment can be put in a form with no division at all,

$$\boxed{U\Delta r \le Kr\Delta U} \tag{11.2}$$

which suits fixed point integer arithmetic and imposes the elasticity requirement on an otherwise standard piecewise linear interest rate model. It belongs in parameter validation rather than in the rate computation, so the hot path is untouched.

The check applies to the curve actually implemented. Replacing an analytic boundary with straight chords can violate its elasticity ceiling, and integer evaluation needs its own rounding headroom. Appendix D gives the chord example and the scope of the numerical checks. A derivative condition for the continuous curve is not by itself a guarantee after token rounding.

12 Conclusion

The condition $\alpha(1+\eta) \leq 1$ separates the supplier's pool share from the rate curve's response to utilization, and turning it into a curve constraint needs exactly one economic input: the protection profile $\bar{\alpha}(U)$. When the condition fails, a supplier's best response is to withdraw until it holds, which raises utilization in the state where the pool can least afford it. Everything else follows from the choice of profile, through the envelope $E = (1-\bar{\alpha})/\bar{\alpha}$, the constraint $\eta \leq E$, and the integral bound of Theorem 2 describing the whole admissible family.

A flat target gives the constant ceiling K, the power law boundary and the endpoint bound K^*. Protecting every fresh depositor gives the profile $1-U$, the hyperbolic boundary and the adjacent rate rule of Equation 7.9. Neither guarantee contains the other, and imposing both takes the smaller envelope.

The guarantee does not set the price level. A compliant floor need not preserve supplier income, while a repair that never lowers rates does, at a cost to borrowers. A protocol adopting the constraint should therefore state its fixed endpoints, allowed kink changes and protection target together. On a piecewise linear curve the check runs at segment endpoints, and a curve with continuous elasticity avoids jumps in the advertised protection at the cost of headroom above K^*. Whether the resulting prices improve participation or market outcomes is a separate question.

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A Boundary Curves and Deposit Examples

The examples in this appendix illustrate the two guarantees without introducing additional assumptions into the main result. Boundary curves describe rate ceilings from a specified anchor, and income comparisons also depend on the rates chosen within those ceilings. Figure A1 draws the boundary curves from a common anchor and the fixed share endpoint bound as the kink moves.

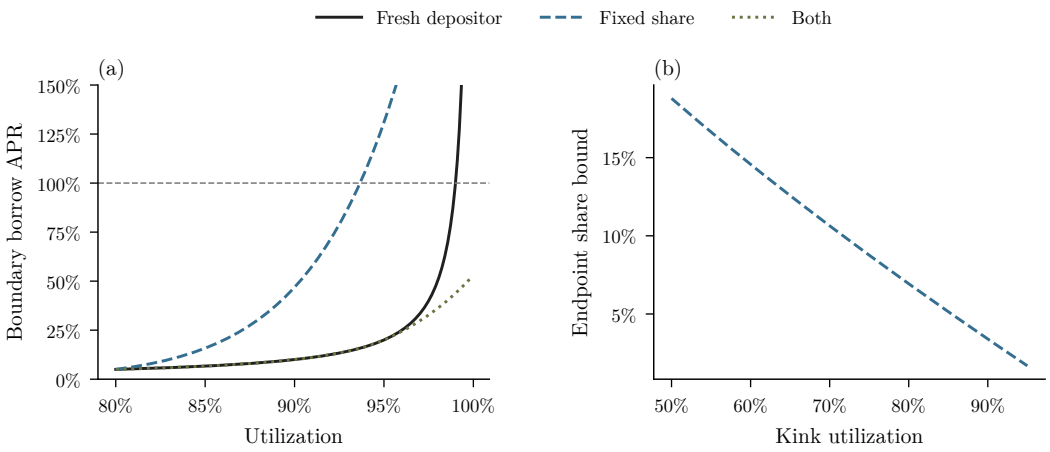


Figure A1: (a) Boundary curves anchored at $(80\%, 5\%)$, with a 100% rate reference. The combined boundary uses the smaller envelope and cannot reach that reference from this anchor. (b) The fixed share endpoint bound as kink utilization varies, with kink APR 5% and maximum APR 100%. These are bounds rather than proposed rate curves.

A.1 A Fresh Depositor Crossing Several Kinks

The following curve satisfies the adjacent rate rule from Section 7 and retains an increasing emergency branch near full utilization. Table 2 checks each segment, and Figure A2 plots the curve.

Table 2: A piecewise linear borrow rate curve satisfying deposit monotonicity.

Segment	Rate ratio	Maximum ratio	Result
$0\% \to 80\%$	1.800	1.800	tight
$80\% \to 90\%$	1.500	1.500	tight
$90\% \to 95\%$	1.481	1.500	valid
$95\% \to 98\%$	1.600	1.600	tight
$98\% \to 99.5\%$	1.750	1.750	tight
$99.5\% \to 99.99\%$	1.964	1.980	valid

Figure A3 shows the guarantee along one deposit path. A fresh depositor arrives at a pool with \$10 million of supply and \$9.99 million of fixed borrowing, and deposits until utilization reaches 85%. The path crosses the kinks at 99.5%, 98%, 95% and 90%. Under Table 2, income rises over the whole path. Under the standard kinked curve of Section 2,

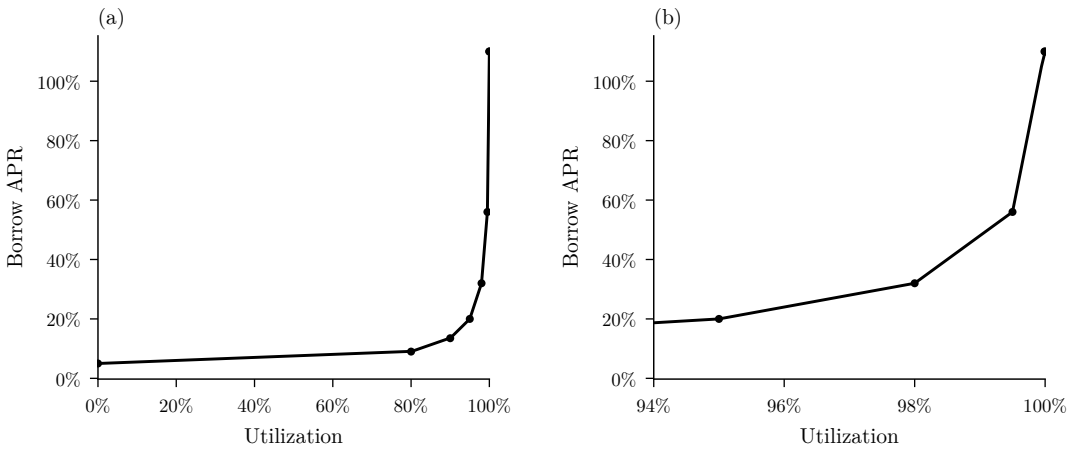


Figure A2: A piecewise linear curve through points at 0%, 80%, 90%, 95%, 98%, 99.5% and 99.99% utilization and borrow APRs of 5%, 9%, 13.5%, 20%, 32%, 56% and 110%, followed by a flat extension to the endpoint at 100% utilization (also marked). Every segment satisfies Equation 7.9, and the nonzero rate at the origin is forced by the base rate bound.

income peaks and then falls, and its marginal dollar turns negative at a deposit of about $1.17 million.

Within each tight linear segment the marginal income approaches zero as the depositor moves towards its left endpoint. It reaches zero there for the most demanding start $u_0 = 1$. For the illustrated start $u_0 = 0.999$, the margin is small but positive. The segment rule is tight for the global guarantee, not necessarily for every individual deposit path.

The curve in Table 2 pays less than the kinked one over this path, ending at about $168,000 against $428,000. This difference reflects the chosen rates rather than an unavoidable cost of the guarantee: at 95% utilization it charges 20% where the rule would allow 100%.

The third curve takes equality in Theorem 4 from a base rate $r_0 = 5\%$, giving $R_b(U) = \min\{r_0/(1-U), 100\%\}$. It charges about 33% at 85% utilization against the standard curve's 29%, reaches its maximum at 95%, and ends the path at about $497,000 a year. It has the same maximum as the standard curve but a higher base. The guarantee constrains percentage sensitivity, not the price level, and does not make every compliant curve earn more.

Below the clamp, the boundary gives

$$\frac{dI}{dx} = \frac{r_0 u^2 (1 - u_0)}{u_0 (1 - u)^2} \tag{A.1}$$

This vanishes when $u_0 = 1$ and is positive for the illustrated start. A boundary means equality in the worst case constraint, not indifference for every depositor.

A.2 An Existing Supplier Under a Power Law Repair

For a supplier who begins above the kink and grows to 5% of final supply, the constrained curve holds both total income and its marginal change nonnegative over the whole deposit path, while the original curve reduces total income over part of it.

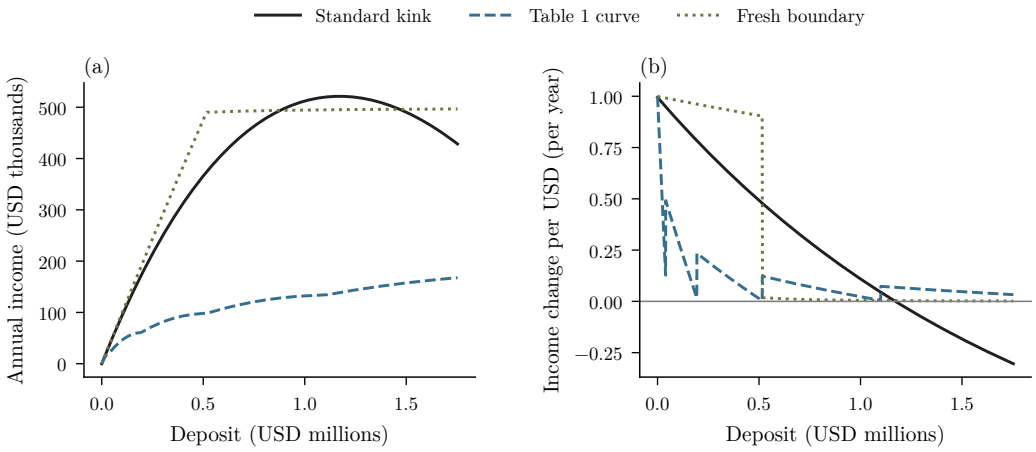


Figure A3: (a) Annualized income and (b) marginal income along the fresh depositor path. The standard kink has zero base rate. Table 2 and the clamped fresh boundary have 5% base rates. The table reaches 110% maximum APR, while the other two reach 100%. These different rate levels are stated explicitly: the comparison illustrates the guarantee, not a common endpoint repair.

What this buys needs stating precisely. This particular repair lies below the original over the replaced interval, so it charges borrowers less and pays suppliers less at the same state. Figure A4 starts with a $100,000 supplier in a $10 million pool at 85% utilization and ends when the supplier owns 5%. The original income eventually falls while the repaired income remains nondecreasing. This is an income direction guarantee, not a claim of higher earnings. Other repairs can raise rates instead, as discussed in Section 9.

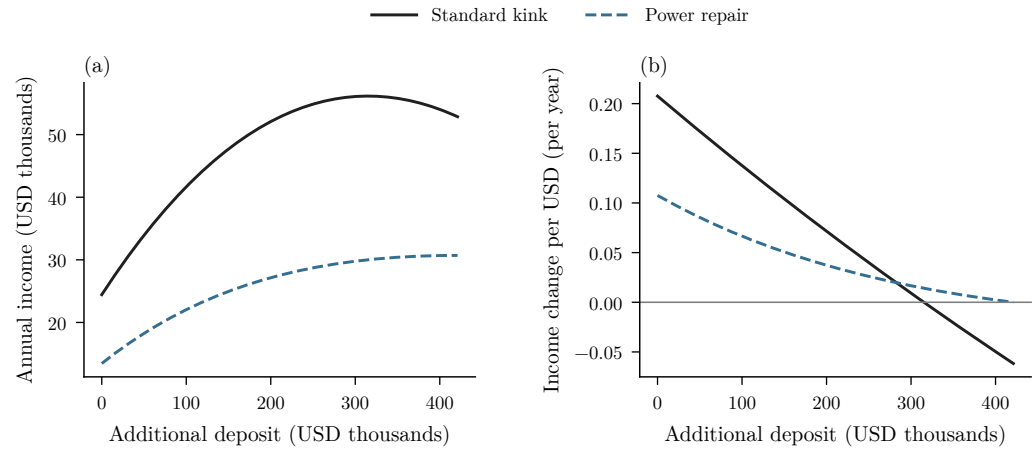


Figure A4: (a) Annualized income and (b) marginal income for the existing supplier, with $8.5 million of fixed borrowing. The path ends at 5% ownership. The repaired marginal income reaches zero at that boundary, while the original becomes negative sooner.

B Constructions for Several Kinks

Assume a multiple kink USDT market with a 0.5% minimum borrow APR, an 80% main kink at 5.25% APR, and a 90% maximum. Below the main kink the rate is $\max\{0.5\%, 5.25\% U/0.8\}$. Above it, straight segments connect the points in Table 3. Figure A5 plots the resulting curve. The parameters are drawn from a deployed configuration. They are not measurements of live market activity or a proposed parameter update. All fixed-kink recursions below use the complete grid: 0%, the exact floor crossing $U = 0.005(0.8)/0.0525 \approx 7.619\%$, and the seven utilizations in Table 3.

Table 3: The illustrative multiple kink rate curve. The floor meets the initial ramp at approximately 7.619% utilization.

Utilization	80%	85%	90%	95%	99%	99.5%	100%
Borrow APR	5.25%	9.4875%	17.9625%	30.675%	47.625%	68.8125%	90%

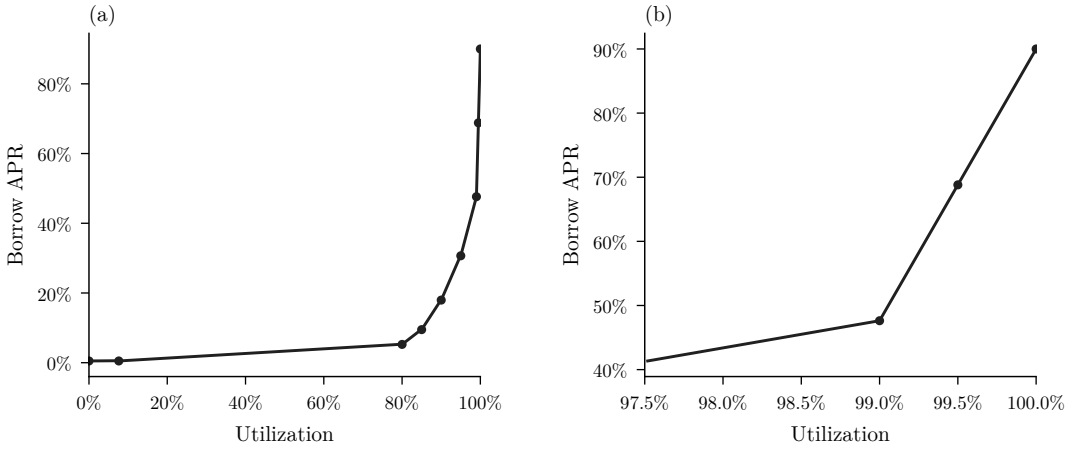


Figure A5: The illustrative multiple kink curve: (a) full utilization range and (b) the final segments. The interval from 99% to 99.5% carries 21.1875 percentage points of borrow APR.

B.1 One Guarantee or Both

Protecting both an incoming depositor and an existing holder up to a fixed share requires the stronger promise at each utilization. The required protected share is therefore the larger profile, and the allowed elasticity the smaller envelope:

$$\bar{\alpha}_{\text{both}}(U) = \max\{\bar{\alpha}, 1 - U\} \qquad E_{\text{both}}(U) = \min\left\{K, \frac{U}{1-U}\right\} \tag{B.1}$$

For $\bar{\alpha} = 5\%$, the fresh depositor rule binds below 95% utilization and the fixed share rule above it. Adding a power law region does not by itself establish both guarantees. Every retained segment must satisfy whichever envelope is tighter there.

On a linear segment $[u_i, u_{i+1}]$, write $\Delta u_i = u_{i+1} - u_i$. The fresh depositor cap on r_{i+1}/r_i is

$$c_i^{\text{fresh}} = 1 + \frac{\Delta u_i}{1 - u_i} \tag{B.2}$$

Checking the fixed share condition at both endpoints gives

$$c_i^{\text{fixed}} = \min\left\{1 + \frac{K\Delta u_i}{u_i},\ \frac{u_{i+1}}{u_{i+1} - K\Delta u_i}\right\} \tag{B.3}$$

where a term imposes no restriction if $u_i = 0$ in the first denominator or the second denominator is nonpositive. Taking $c_i = \min\{c_i^{\text{fresh}}, c_i^{\text{fixed}}\}$ imposes both guarantees, with the complete segment constraint $r_i \le r_{i+1} \le c_i r_i$.

B.2 Rate Floors and Repair Choices

If a market must still reach its existing maximum, limiting the increase on each segment places a lower bound on the rates leading up to it. For fixed kinks and final rate r_n, this bound follows by setting $\ell_n = r_n$ and working backwards:

$$\ell_i = \frac{\ell_{i+1}}{c_i} \tag{B.4}$$

An unbounded cap gives a zero lower bound. Every feasible curve with these kinks and top rate must lie above the resulting floor, but moving all rates down to it can reduce supplier income even where the baseline was already compliant. A protocol that wants to preserve that income can instead require each repaired rate to remain at least as high as its baseline value.

Writing those baseline rates as r_i^0, the least such repair is

$$r_n^+ = r_n^0, \qquad r_i^+ = \max\left\{r_i^0, \frac{r_{i+1}^+}{c_i}\right\} \tag{B.5}$$

Because the baseline is nondecreasing, the backwards recursion satisfies both segment inequalities. Any admissible curve above the baseline must meet the same backwards bounds, making this the pointwise smallest repair that preserves existing rates. Supplier income cannot fall under an unchanged market history, although the higher rates increase borrower costs.

The opposite choice fixes the base and never raises a baseline rate:

$$r_0^- = r_0^0, \qquad r_{i+1}^- = \min\{r_{i+1}^0, c_i r_i^-\} \tag{B.6}$$

This is the greatest admissible curve below the baseline, but its top rate is not fixed. Keeping the original top while never raising a rate is infeasible whenever the baseline falls below the corresponding fixed top floor.

Allowing both increases and decreases gives a third pricing choice. The mixed repair in the figures minimizes $\sum_i w_i(r_i - r_i^0)^2$ over feasible kink rates with the top held fixed, using trapezoidal utilization weights w_i. This objective measures changes to the kink parameters. It does not optimize borrower or supplier welfare, nor is it the exact integral of squared curve error.

For the fresh depositor rule, the fixed top floor has base APR 3.47%, while preserving every baseline rate requires 6.39%. Under both guarantees the minimum base is 8.68% on these kinks, and the floor, raise only and mixed repairs coincide. The numbers describe this kink grid and are not universal minimum prices for monotonicity. A repair that never raises the baseline instead reaches only 12.04% maximum APR under the fresh rule and 4.82% under both, rather than 90%. Figure A6 compares the two directions under both guarantees.

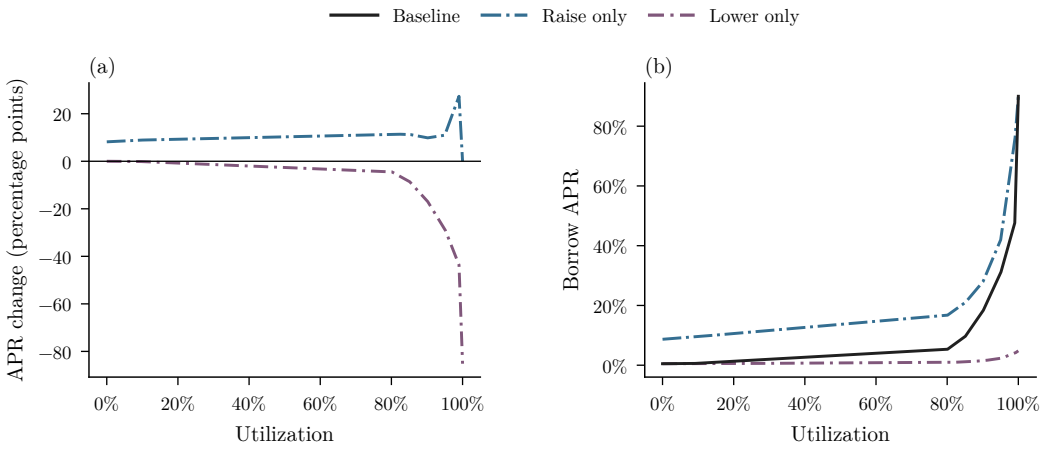


Figure A6: Both guarantees: (a) APR changes from baseline and (b) resulting rates. Raise only retains the 90% top and coincides with the floor and mixed repair. Lower only retains the original base and never raises a rate, but must lower the maximum. The different endpoint choices are part of the comparison.

B.3 Power Law Regions on a Multiple Kink Curve

With several kinks, a rate lowering power branch may fail to catch the original curve before full utilization even when the main kink and maximum satisfy the global endpoint bound. Retaining low rates at intermediate kinks can leave too little utilization range for the final increase, so preserving the maximum may require raising rates before the offending segment.

For the constructions in this subsection, assume a continuous, nondecreasing piecewise linear baseline with strictly positive base rate, as in the illustrative curve. This ensures the running-infimum minorant remains positive. For the fixed-share envelope, a baseline with a positive-slope first segment through the origin is also permitted when $K \geq 1$. When $K < 1$, its running-infimum minorant is identically zero and is outside the positive-rate class.

These replacement regions can be constructed by measuring the baseline relative to the fastest rate increase allowed by the guarantee. Let F be a positive boundary function on $U > 0$, with $F'/F = E(U)/U$, and write $q_0(U) = R_0(U)/F(U)$. Admissibility requires R/F to be nonincreasing: once the rate has fallen behind its boundary, it cannot catch up by exceeding the envelope later. The largest admissible curve below the baseline and the smallest one above it are therefore

$$\boxed{\begin{aligned} R_-(U) &= F(U) \inf_{0<v\leq U} q_0(v) \\ R_+(U) &= F(U) \sup_{U\leq v\leq 1} q_0(v) \end{aligned}} \tag{B.7}$$

Values at zero are taken by continuity. The first construction preserves as much borrower pricing as possible without raising any rate. The second preserves supplier pricing by raising only what the constraint requires.

Proof. The envelope condition is equivalent to R/F being nonincreasing. Any admissible minorant therefore satisfies $R(U)/F(U) \leq q_0(v)$ for every $v \leq U$, giving the first bound.

The running infimum is itself nonincreasing and meets that bound. The majorant follows by the same argument with future points and a running supremum. On a piecewise linear nondecreasing baseline, the resulting curve follows either the baseline or a constant multiple of F. Both are nondecreasing, and the construction is continuous at contacts. Thus the bounds also belong to the rate class assumed here. □

For fixed share protection, $F(U) = U^K$, so the replacement intervals are power laws. On the illustrative multiple kink curve with $K = 19$, the minorant first departs at 99% utilization and reaches only 57.65% APR at full utilization. It cannot retain the 90% maximum. The majorant retains that maximum by raising rates earlier, while still preserving the original base and 80% kink rate. Figure A7 shows both curves and their elasticity. The formulas include interior extrema of q_0 on each original segment, since checking the values at the kinks alone is not sufficient for the running supremum.

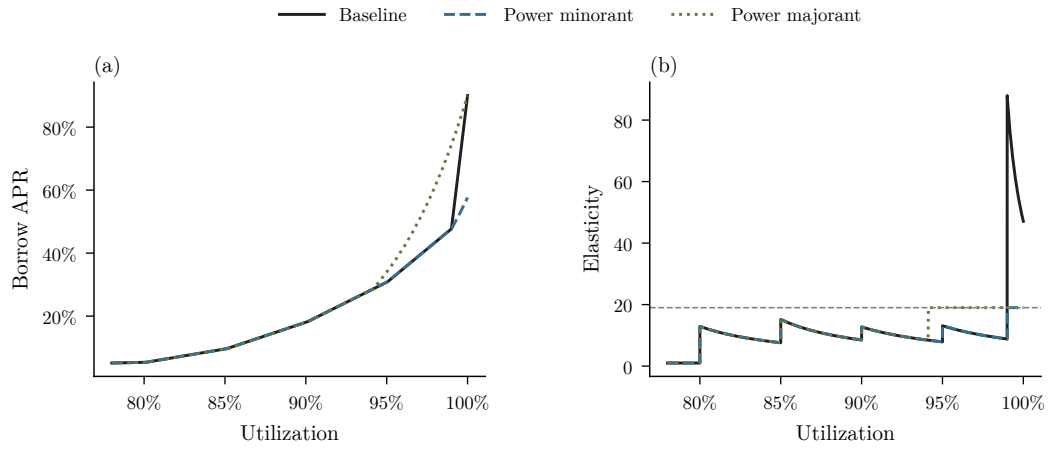


Figure A7: Analytic fixed share repairs with $K = 19$: (a) borrow APR and (b) elasticity. The power minorant never raises rates but misses the original maximum. The power majorant preserves the maximum and raises the rates needed to reach it. The horizontal line marks the elasticity ceiling.

To impose both guarantees, let $t = K/(K + 1)$ and normalize the combined boundary at the origin:

$$F(U) = \begin{cases} (1 - U)^{-1}, & 0 \le U \le t \\ (K + 1)(U/t)^K, & t < U \le 1 \end{cases} \tag{B.8}$$

The same construction then uses hyperbolic regions below t and power law regions above it, retaining the baseline wherever contact permits. This is distinct from the fixed kink linear implementation. The combined analytic majorant has base APR approximately 1.85% and keeps the 90% maximum, compared with 8.68% for the coarse fixed kink repair. Figure A8 compares the two.

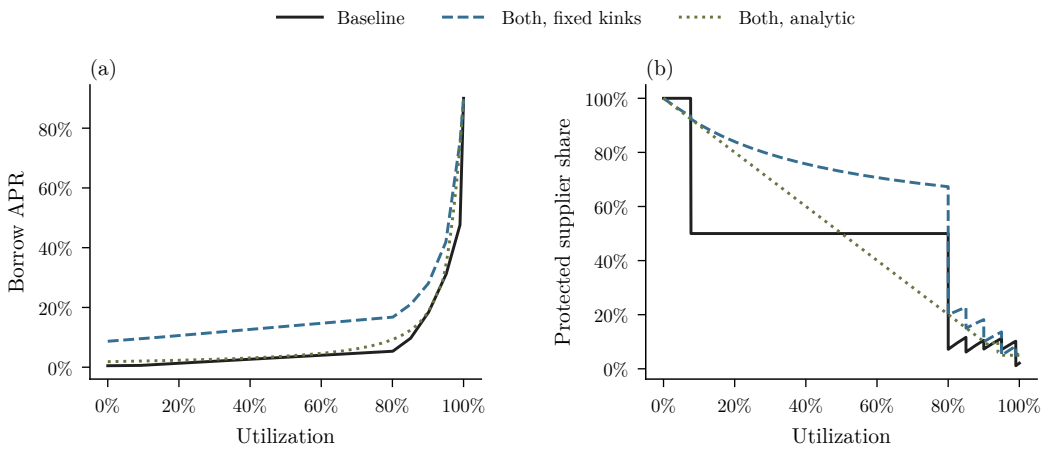


Figure A8: Both guarantees, with 5% fixed share protection. The fixed kink and analytic curves are the least repairs nowhere below the same baseline in their respective classes. (a) Borrow APR. (b) Actual protected share. The analytic curve can place new contacts and use curved regions, so it is not restricted to the original kink grid.

B.4 The Cost of Keeping the Kinks

The combined envelope has a finite budget even though the fresh depositor envelope alone does not. With no restriction on kink placement, its fixed top base bound is

$$\boxed{R_b(0) \geq \frac{R_{\max}}{F(1)} = \frac{R_{\max} t^K}{K+1}} \tag{B.9}$$

For $K = 19$ and $R_{\max} = 90\%$, this is 1.70%. It is attained by a scaled combined boundary. That curve need not lie above the baseline. Preserving the baseline raises the bound to the 1.85% found above.

Figure A9 subdivides each original segment and recalculates the linear floor. At 64 subsegments per original segment, its combined base falls from 8.68% to 1.80%, approaching the continuous bound. Under the fresh rule alone it falls towards zero as more of the rate increase can be concentrated near full utilization. A finite set of kinks and the continuous admissible family are different design spaces, and their bounds should not be interchanged.

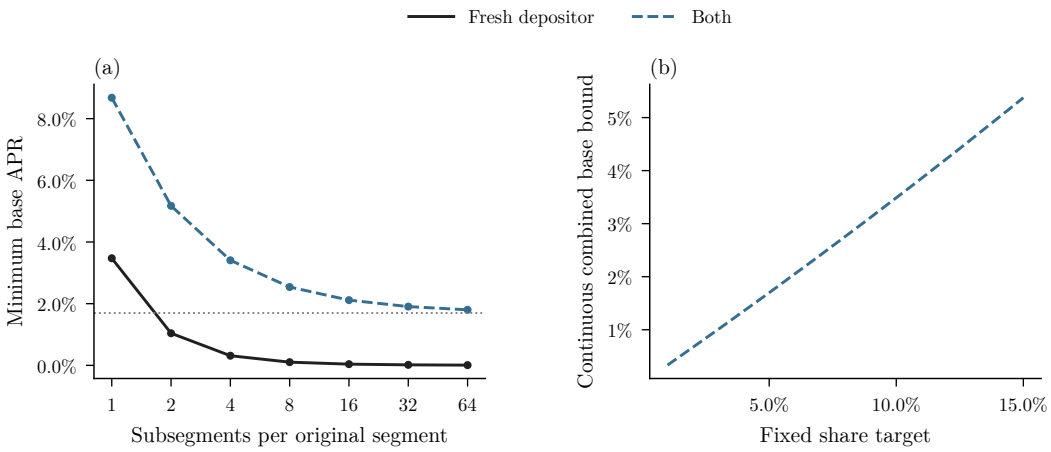


Figure A9: (a) Minimum base APR at a fixed 90% maximum as every original interval is subdivided equally. The dotted reference is the continuous combined bound, not a floor for the fresh only rule. (b) The continuous combined base bound as the fixed share target varies. These floors do not additionally require staying above the baseline.

C Additional Interest Scenarios

The histories below extend Section 10. They use the same constant 5% reserve factor, pay interest out rather than compounding it, and prescribe identical principal and utilization paths for each rate curve.

Under both guarantees on the fixed kinks, the three repairs coincide and pay approximately 3,165 USDT on the deposit history specified in Section 10. A second example begins with a 10,000 USDT depositor in a pool holding 1 million USDT against 990,000 borrowed. The depositor adds 20,000 on each of days 60, 120, 180, 240 and 300, while borrowing remains fixed. Annual interest is about 19,667 USDT under the combined repair and 13,453 under the baseline. Figure A10 shows both histories. Ownership eventually exceeds 5%, so the local guarantee still has to be read from the protection profile rather than inferred from the label of the curve.

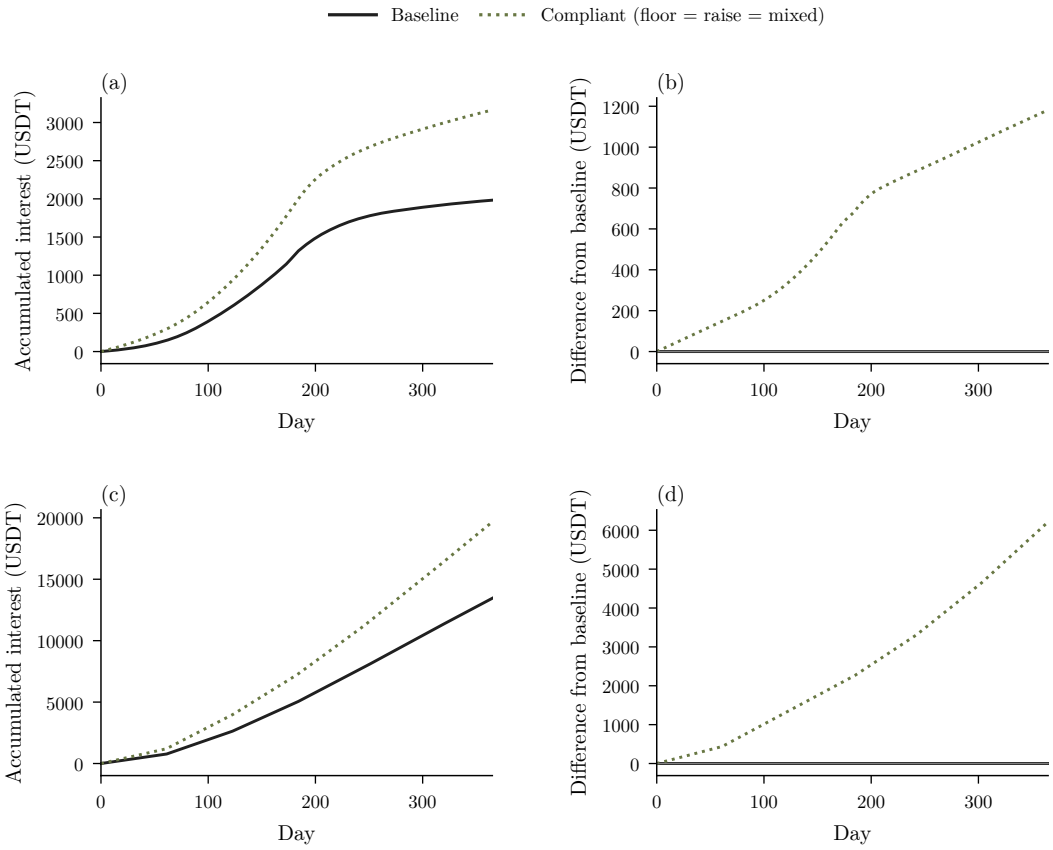


Figure A10: Both guarantees on fixed kinks. Top row: the single deposit history. Bottom row: repeated additions into fixed borrowing. Left: accumulated interest. Right: difference from baseline. All curves use identical principal and utilization histories, a 5% reserve factor and no compounding. Exit at the end returns principal separately.

As in Section 10, higher supplier income on an unchanged history is funded by borrowers. Figure A11 follows this transfer as the fixed share target varies, retaining the fresh guarantee, original kinks and maximum. It measures the resulting prices rather than their effect on welfare.

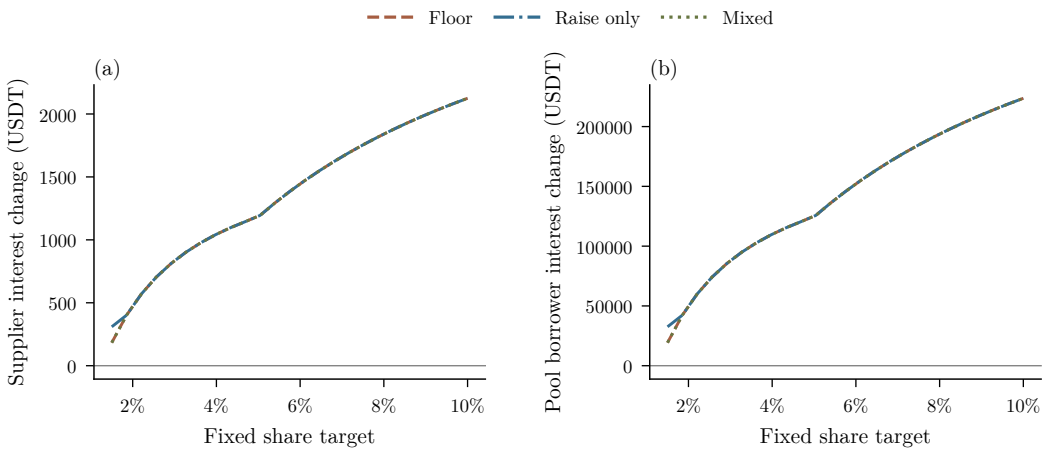


Figure A11: Both guarantees as the fixed share target varies. (a) Annual supplier interest change and (b) aggregate borrower interest change, on the single deposit history. The repairs coincide over much of this range. With fixed 1% ownership and a 5% reserve factor, panel (a) is 0.0095 times panel (b).

To test dependence on utilization exposure, Figure A12 allocates a year between 80% and 99% utilization for the same 10,000 USDT principal. The fresh only floor can gain at low utilization and lose at high utilization, whereas the raise only repair cannot lose under any such mix. The analytic power minorant and majorant have the corresponding sign restrictions. These comparisons distinguish conclusions that follow from the direction of rate changes from outcomes that depend on a chosen history.

The sign condition holds independently of these examples, while the accumulated earnings depend on their specified histories. Predicting borrowing demand, liquidity, utilization or default risk after repricing requires a separate model. At the limiting case of 100% ownership, supplier income equals total borrower interest, so any strictly increasing rate curve makes added supply reduce income at fixed borrowing, as shown in Appendix F.

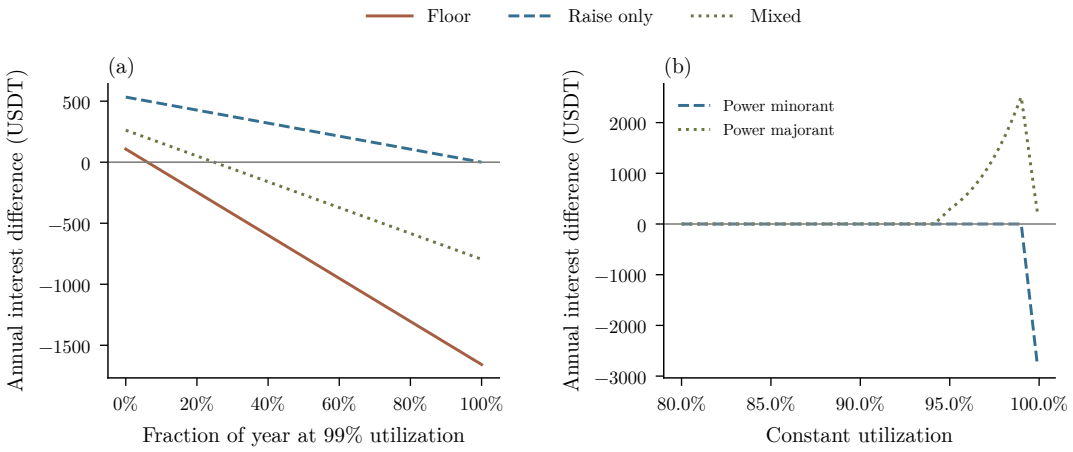


Figure A12: Sensitivity to utilization exposure. (a) Fresh only fixed kink repairs: a year is split between 80% and 99% utilization, with the high utilization fraction on the horizontal axis. The top legend applies to this panel. (b) Fixed share analytic power repairs: annual interest differences at a constant utilization, with a separate legend. Both panels hold principal at 10,000 USDT and use a 5% reserve factor.

D Discretization and Numerical Checks

One caveat applies to any discretization of a smooth boundary. Sampling the constant elasticity boundary at two points replaces it by a chord whose slope is the average of the slopes it spans, so the chord is steeper than the boundary at the segment's left endpoint. That endpoint is where elasticity binds, since the chord of a convex curve has negative intercept and η therefore decreases across the segment, and the sampled curve is more elastic there than the boundary it was drawn from. Sampling $R_o(U/O)^{19}$ at 80% and 90%, for instance, gives a segment whose elasticity at 80% is about 67. Figure A13 shows this chord. A ceiling imposed on a sampled curve has to be chosen with considerable headroom, or the kinks placed close enough together that the chords track the boundary.

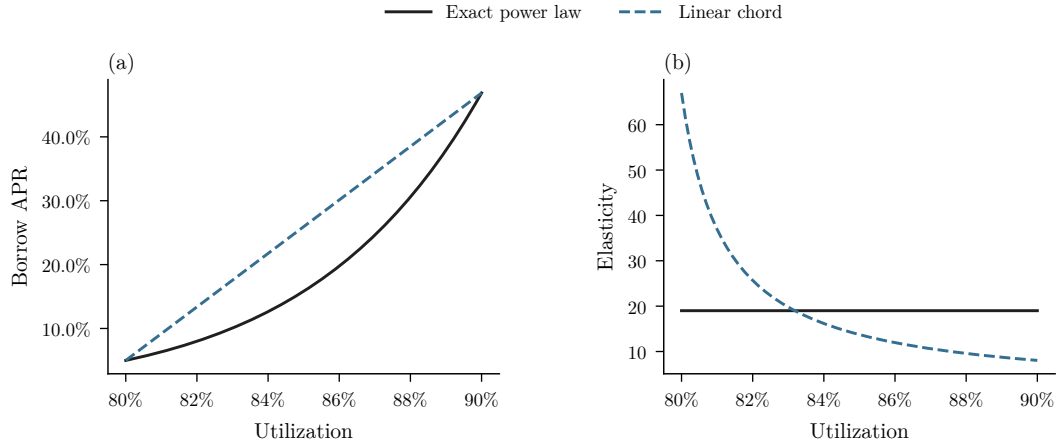


Figure A13: (a) A constant elasticity power law and its chord between 80% and 90% utilization. (b) Their elasticities. Matching endpoint rates does not preserve the ceiling between the endpoints.

The checks above concern the continuous rate represented by the parameters. Integer evaluation introduces small steps, for which a derivative is not defined. In the illustrative multiple kink calculation, evaluating the integer formula at all 1,000,001 utilization ticks differs from the rational curve by less than 10^{-6} in decimal APR. That agreement is not a proof of exact transaction level monotonicity after all token and interest rounding. An implementation needs rounding headroom and finite increment checks at its own precision, and a shared global weight table may not express an arbitrary per market repair.

E Smooth Constructions

A protocol may also want the advertised protected share to vary continuously with utilization. This requires a continuous elasticity profile and imposes an additional cost beyond the deposit monotonicity constraint.

The minimal change construction is continuous in the rate but not in its derivative: below the kink the slope is R_o/O while the power branch leaves at KR_o/O, so unless $K = 1$ the two disagree, and the same mismatch occurs in reverse at U_c. That is worth something more than tidiness, because the protected share is computed from elasticity,

$$\alpha_{\text{safe}}(U) = \frac{1}{1 + \eta(U)} \tag{E.1}$$

and therefore inherits every discontinuity of it. Under the minimal change curve α_{safe} falls from $\frac{1}{2}$ to $1/(1+K)$ at a single point, so a supplier holding 30% of the pool is protected at $U = O - \varepsilon$ and unprotected at $U = O + \varepsilon$, and the guarantee the protocol advertises jumps while the state variable it depends on moves continuously. Figure A14 shows the jump next to the smooth branch built below.

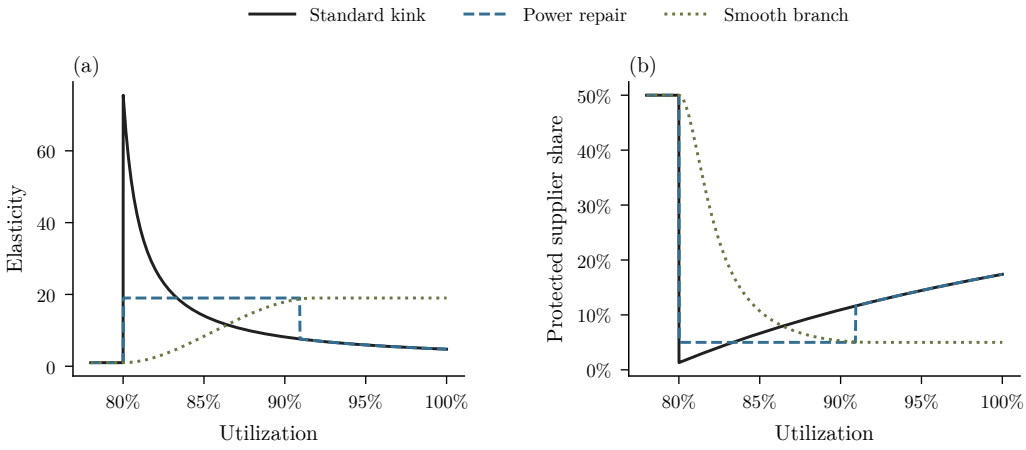


Figure A14: (a) Elasticity and (b) protected share for the standard kink, the minimal power repair and a smooth branch. The repaired curves satisfy $K = 19$, but the standard kink does not. The power repair jumps in protected share, while the smooth branch changes continuously.

The integral of Theorem 2 also builds curves instead of only bounding them. If η is continuous on $[U_a, U_b]$ then

$$R_b(U) = R_a \exp\left(\int_{U_a}^{U} \frac{\eta(v)}{v}\, dv\right) \tag{E.2}$$

is positive, continuously differentiable, and has elasticity exactly η. Conversely every positive C^1 curve through (U_a, R_a) is of this form with η its own elasticity, and a piecewise continuous η gives the same correspondence away from its jumps. Prescribing elasticity is therefore equivalent to prescribing the rate, and the construction below works in that direction, because it makes the protected share continuous.

Removing the jump means asking elasticity itself to vary continuously. Choose a continuous profile η on $[O, 1]$ that starts at the initial value $\eta(O) = \eta_0$, stays below a ceiling K, and recover the rate from it,

$$R_b(U) = R_o \exp\left(\int_O^U \frac{\eta(v)}{v}\, dv\right) \tag{E.3}$$

Then $R_b' = \eta R_b/U$ is continuous, the branch is C^1, and matching η_0 at the kink makes the assembled curve C^1 across the join. Appendix E.1 carries out the construction explicitly for one such family.

The ceiling cannot be taken all the way down to the endpoint bound, however.

Theorem 5 (C^1 makes K^* unattainable)**.** *Let $R_b \in C^1([O, 1])$ be positive with $R_b(O) = R_o$ and $R_b(1) = R_{\max}$, and let its elasticity satisfy $\eta(O) = \eta_0 < K^*$. Then*

$$\boxed{\sup_{U \in [O,1]} \eta(U) > K^*} \tag{E.4}$$

so a ceiling $K = K^*$ is infeasible under C^1, while every $K > K^*$ remains feasible.

Proof. In $x = \ln(U/O)$ with $L = \ln(1/O)$, Equation 6.3 reads $\int_0^L \eta\, dx = K^* L$, so K^* is the average elasticity the curve must achieve. Continuity with $\eta(0) = \eta_0 < K^*$ holds η below K^* over an interval of positive length, and an average of K^* then forces $\eta > K^*$ somewhere later. Feasibility for $K > K^*$ follows from the construction in Appendix E.1. □

Theorem 3 makes K^* the minimum worst case elasticity and Theorem 5 makes it a minimum that is approached without being reached, since the curve attaining it is the pure power law, which is exactly the one that gives up C^1 at the kink.

The design choice is therefore a plain one. The power repair is the greatest admissible curve below the single kink baseline and is C^0, with a protected share that drops at the kink. A continuous elasticity profile gives C^1 and a protected share that slides, but only by spending elasticity headroom above K^*: for the illustrative inputs, a ceiling of $K = 19$ buys a transition spread from 80% to 91.9% utilization. Both leave O, R_o and $R_{\max}$ where they were.

E.1 An Explicit C^1 Branch

For a concrete construction, we use the following family to compute the headroom required by smoothing.

Work in $x = \ln(U/O) \in [0, L]$ with $L = \ln(1/O)$ and $A = \ln(R_{\max}/R_o) = K^* L$, and let $s(t) = 3t^2 - 2t^3$ be the standard smoothstep, which has $s(0) = 0$, $s(1) = 1$, $s'(0) = s'(1) = 0$ and $\int_0^1 s = \frac{1}{2}$. For $0 < w \le L$, prescribe elasticity as a ramp that finishes within the interval, followed by a plateau,

$$\boxed{\eta(x) = \begin{cases} \eta_0 + (\eta_{\max} - \eta_0)\, s(x/w), & 0 \le x \le w \\ \eta_{\max}, & w < x \le L \end{cases}} \tag{E.5}$$

and recover R_b from it by the integral construction. Taking $\eta_0 = 1$ matches the ramp below the kink, so the curve is C^1 there, and the vanishing endpoint slopes of s make η itself C^1 and hence $R_b \in C^2$.

The plateau is not a free parameter, since the branch must still arrive at $R_{\max}$.

Proposition 6 (The cost of smoothing)**.** *The endpoint condition $\int_0^L \eta\, dx = A$ determines the plateau uniquely,*

$$\boxed{\eta_{\max}(w) = \frac{A - \eta_0 w/2}{L - w/2} \qquad 0 \le w \le L} \tag{E.6}$$

with $\eta_{\max}(0) = K^*$ and, whenever $\eta_0 < K^*$,

$$\frac{d\eta_{\max}}{dw} = \frac{L(K^* - \eta_0)}{2\,(L - w/2)^2} > 0 \tag{E.7}$$

so $\eta_{\max}(w) > K^*$ for every $w > 0$ in this domain. The value at $w = 0$ denotes the nonsmooth limit. For a width in $[0, L]$, the branch respects a ceiling $K > \eta_0$ if and only if

$$\boxed{w \le \frac{2L\,(K - K^*)}{K - \eta_0}} \tag{E.8}$$

Proof. Integrating the profile, $\int_0^w \eta\, dx = \eta_0 w + (\eta_{\max} - \eta_0) w \int_0^1 s = w(\eta_0 + \eta_{\max})/2$, so

$$\int_0^L \eta\, dx = \frac{w(\eta_0 + \eta_{\max})}{2} + \eta_{\max}(L - w) = \eta_{\max}\left(L - \frac{w}{2}\right) + \frac{\eta_0 w}{2} \tag{E.9}$$

Setting this equal to A and solving gives $\eta_{\max}$, and differentiating gives the stated derivative, whose sign is that of $K^* - \eta_0$ because $A = K^* L$. Imposing $\eta_{\max}(w) \le K$ and rearranging yields the width bound. □

Proposition 6 is the quantitative form of Theorem 5: the plateau meets K^* only in the degenerate case $w = 0$, which is the power law and is not C^1. The rate is available in closed form as well, with $\int_0^t s = t^3 - t^4/2$,

$$R_b(U) = R_o \exp H(x) \qquad H(x) = \begin{cases} \eta_0 x + (\eta_{\max} - \eta_0)\, w \left[t^3 - \frac{t^4}{2}\right]_{t = x/w}, & x \le w \\ \dfrac{w(\eta_0 + \eta_{\max})}{2} + \eta_{\max}(x - w), & x > w \end{cases} \tag{E.10}$$

so evaluating it requires no quadrature.

Table 4 records the trade for the illustrative inputs, where $L = 0.2231$, $A = 2.9957$ and $K^* = 13.43$. A wider ramp buys a gentler transition and costs protected share, and at $w = L$ the ramp fills the whole branch, the plateau is reached only at full utilization, and $\eta_{\max} = 2K^* - \eta_0$, which equals $2K^* - 1$ under the standing choice $\eta_0 = 1$. For a 5% target, meaning $K = 19$, the widest admissible ramp has $w = 0.1382$ and carries elasticity from 1 to 19 between 80% and 91.9% utilization while still reaching $R_{\max}$ exactly at $U = 1$. Figure A15 plots the same trade over all widths.

Table 4: Smoothing width against plateau elasticity, for $O = 80\%$, $R_o = 5\%$, $R_{\max} = 100\%$ and $\eta_0 = 1$.

Width w	Ramp ends at	$\eta_{\max}$	Protected share
0.0000	80.00%	13.43	6.93%
0.0500	84.10%	14.99	6.25%
0.1000	88.41%	17.01	5.55%
0.1382	91.86%	19.00	5.00%
0.2231	100.00%	25.85	3.72%

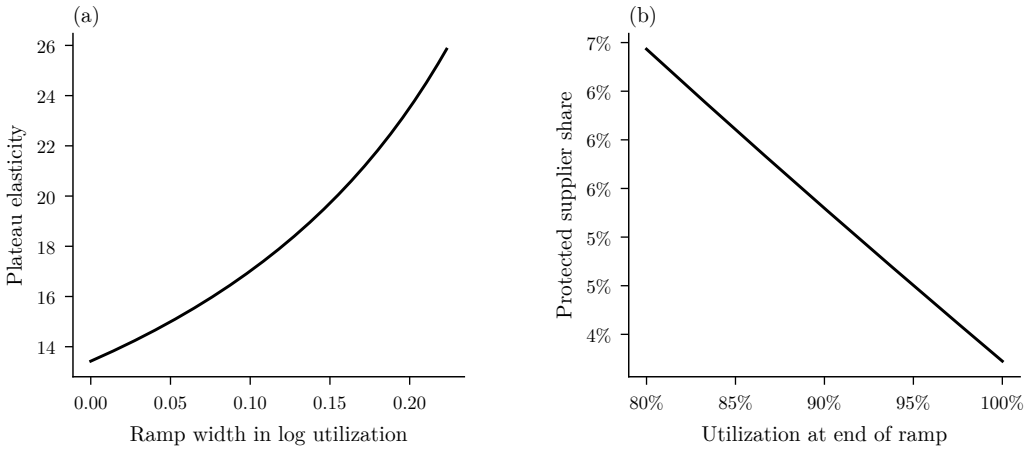


Figure A15: (a) Plateau elasticity against ramp width, with $0 \leq w \leq L$. (b) The corresponding protected share against the utilization where the ramp ends. The value K^* is reached only in the nonsmooth limit $w = 0$.

F The Full Supplier Limit

Setting $\alpha = 1$ gives $\eta \leq 0$. When one supplier owns all deposits, $TU = B$ and their income is $I = BR_b(U)$. Depositing while B remains fixed reduces utilization and therefore income wherever the borrow rate is increasing. Equivalently, with $\alpha = 1$ Equation 4.4 becomes

$$\frac{dI}{d\delta} = -U^2 R_b'(U) \tag{F.1}$$

No strictly increasing rate curve can protect this supplier. A flat rate gives equality. Figure A16 uses the single kink example with a 1% floor, a supplier initially holding all \$10 million of supply and \$9.5 million of fixed borrowing. The plotted interval remains on rising portions of the curve, and a sufficiently large deposit would eventually reach the flat floor.

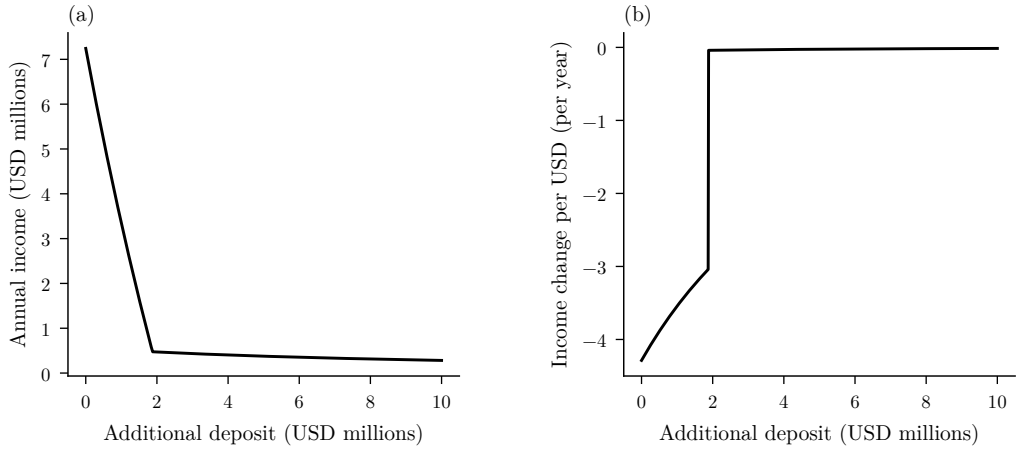


Figure A16: A supplier owning the whole pool: (a) annualized income and (b) its marginal change under additional deposits. The guarantee cannot hold on a rising rate branch.